MILITARY SPECIFICATION

EDGESEAL MATERIALS, CHEMICAL-BIOLOGICAL FILTER

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of heat-fusing material for edgesealing fibrous components of protective equipment.

1.2 Classification. Edgeseal material shall be one of the following types as specified (see 6.2).

Type I - For filter element sets in M17, M17A1 and M17A2 chemical-biological masks.
Type II - For filter panels in M2 infant protector.

2* APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4240

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SPECIFICATIONS

MILITARY

MIL-F-51425 - Filter Element Set, Chemical-Biological Mask, M13A2.

STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 406 - Plastics; Methods of Testing.
Fed. Std. No. 595 - Colors.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D412 - Rubber Properties in Tension, Test for.
D624 - Rubber Property - Tear Resistance, Test for.
D1203 - Loss of Plasticizer from Plastics (Activated Carbon Methods), Test for.
D2137 - Rubber Property - Brittleness Point of Flexible Polymers and Coated Fabrics, Test for.

(Application for copies of the above publications should be addressed to ASTM, 1916 Race Street, Philadelphia, 'PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I. Type I edgeseal material shall be "Wooster A-466-2 or Providence D-3557 Low Migration Plastisol" (see 6.5). The material shall consist of polyvinyl chloride (PVC) dispersed in suitable liquid plasticizers, with sufficient stabilizers, and with pigments and fillers yielding a gray color similar to No. 36081 of Fed. Std. No. 595. The material shall not be

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hazardous or detrimental to molding equipment. The material shall be packaged in containers whose normal capacity is not greater than 5 gallons (U.S*). Any change in ingredients or in quantities beyond the tolerable range of variation (see (b) below) shall constitute cause for submission of a new preproduction sample (see 3.8). The contractor shall submit the following information .which he may designate as "company confidential", in which case it will be so treated by the Government:

(a) Chemical and trade names of all ingredients in the edgeseal material. If the chemical name can not be obtained, the trade. name shal be sufficient.

(b) The purity and quantity, with allowable tolerances for each ingredient as required to fully define the edgeseal material. Quantity of each ingredient shall be expressed in percent by weight or in parts by weight.

3.1.2 Type II. Type II edgeseal material shall be made from either one of the following formulations. In all other respects type II edgeseal material shall satisfy 3.1.1.

(a) Wooster A466-2A low migration plastisol (see 6.5) consisting of A466-2 with a minimum of 88 percent of the approved PVC base resin and maximum of 12 percent of a substitute of PCV resin.

(b) Providence D-3558 low migration plastisol (see 6.5) consisting of D-3557 with a minimum of 88 percent of the approved PVC base resin and a maximum of 12 percent of a substitute of PVC resin.

3.2 Compatibility (type I only). Filter elements molded with type I edgeseal material shall meet the requirements of MIL-F-51425 when inspected as specified in 4.3.2.2.

3.3 Physical properties. Standard specimens fabricated from the edgeseal material shall conform to the requirements of table I before and after aging.

TABLE I. Physical properties.

Property	Requirement		Test paragraph
	Unaged	Aged*	
Hardness (shore A)	65 + 5	65 + 5	4.4.4.2
Average tensile strength (lb/in2)	1000 min	1000 min	4.4.4.3
Degree of blocking, max	Slight	---	4*4.4.4
Average tear strength (lb/in)	100 min	100 min	4.4.4*5

*Aging. The test specimens shall be subjected to an exposure cycle of 24 hours at 175° + 2°F and 24 hours at -40° + 2°F. The specimens shall be held at 73.5° + 2°F and 50 + 2 percent relative humidity for a minimum of 48 hours before testing.

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3.4 Brittleness. Specimens fabricated from the edgeseal material shall show no cracks, fissures, crazing, or ruptures when tested as specified in 4.4.4.6 at a temperature of -10° + 2 F.

3.5 Volatile loss. The edgeseal material shall not show a loss in weight greater than 0.5 percent when tested as specified in 4.4.4.7.

3.6 Viscosity. The viscosity of the edgeseal material shall be 4000 ± 1000 centipoises when tested as specified in 4.4.4.8.

3.7 Instruction sheet. The preproduction lot shall be accompanied by an instruction sheet containing the following information:

(a) Title, number, date of this specification and type of edgeseal material.
(b) Materials designation.
(c) Recommended storage conditions
(d) Estimated shelf life (under recommended storage conditions) to maintain required viscosity (see 3.6).
(e) Other information considered useful by the manufacturer or specified by the procuring activity.

3.8 Preproduction. Prior to the start of regular production, a preproduction lot of edgeseal material shall be produced for testing in accordance with 4.3.

4 l QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence including actual test results acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction inspection.

4.3.1 Sample.

4.3.1.1 Edgeseal material. A preproduction sample of 55 gallons of edgeseal material shall be produced using the same methods, materials, and equipment as will be used in regular production.

4.3.1.2 Filter elements (type I only). A preproduction lot of filter elements shall be produced in accordance with the preprocluction provisions of MIL-F-51425.

4.3.2 Inspection procedure.

4.3.2.1 Edgeseal material. A l-quart specimen shall be taken from the preproduction sample and test slabs shall be prepared as specified in 4.4.4.1 for testing in accordance with 4.4.4.2 through 4.4.4.7. In addition, a portion of the l-quart specimen shall be tested for viscosity in accordance with 4.4.4.8.

4,3.2.2 Filter elements (type I only). The preproduction lot shall be examined and tested in accordance with the preproduction provisions of MIL-F-51425 to deter-mine compliance with the compatibility requirement specified in 3.2 (see 6.6).

4.3.3 Acceptance/rejection criteria. If the edgeseal material fails any of the tests specified in 4.3.2 it shall be rejected.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the edgeseal material of one type produced by one manufacturer in no more than 24 consecutive hours under essentially the same manufacturing conditions and with no change in materials, provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.3)-

4.4.2 Sampling. In the case of a continuous process, the unit of product shall be 5 gallons. A sample of five units of product shall be taken at random from each lot. From each container in the sample, a l-quart specimen shall be removed and placed in a clean, dry container labeled to identify the specimen with lot and sample container represented* In the case of batch processing, a 5-gallon sample shall be taken from each batch while the material is stored in the final mixing tank. From the 5-gallon sample a l-quart specimen shall be removed and placed in a clean, dry container labeled to identify the specimen with batch represented.

4.4.3 Inspection procedure. Test slabs shall be prepared from each l-quart specimen as specified in 4.4.4.1, and the slabs shall be tested in accordance with 4.4.4.2 through 4.4.4.7. Also, a portion of each 1-quart specimen shall be tested as specified in 4.4.4.8.

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4.4.4 Tests.

4.4.4.1 Preparation of test specimens. Specimens for testing the fused edge-seal material shall be made in open aluminum molds 6-1/2 inches by 6-1/2 inches by no more than 1/2 inch thick, with a cavity size 6 inches by 6 inches and of sufficient depth to produce a sheet 0.075 -t- 0.005 inch in thickness. The mold cavity shall be filled with fully dispersed edgeseal material and placed in a mechanical, convection, recirculating type oven at a temperature of 375° + 2°F so that the mold is essentially level. The edgeseal material shall be fused at 350°F as indicated by a thermocouple placed in the center of the thickness of the sample. (A suitably calibrated thermocouple shall be used to determine the internal. temperature of the material; care should be exercised to insure that the thermocouple does not bottom on the mold cavity.) Once 350 F is indicated by the thermocouple, hold specimen in oven for 3 (plus 1 or minus O) minutes, remove the mold from the oven, and cool it at room temperature for 8.0 + 1.0 minute followed by force-cooling to room temperature within 10.0 + 1.0 minute of removal from oven. Conditioning of specimens shall commence within 48 hours of molding the fused sheets. The conditioning shall be in accordance with the applicable test method.

4.4.4.2 Hardness. Three specimens prepared as specified in 4.4.4.1 shall be tested in accordance with method 1082 of Fed. Std. No. 406.

4.4.4.3 Tensile strength. Six specimens prepared as specified in 4.4.4.1 shall be tested in accordance with ASTM D412. The specimens shall be cut with standard dumbell die C.

4.4.4.4 Blocking. Six specimens prepared as specified in 4.4.4.1 shall be cut and formed into three sandwiches (two specimens per sandwich) and tested in accordance with method 1131 of Fed. Std. No. 406. A pressure of 1 pound per square inch and a temperature of 158° ± 2°F shall be used.

4.4.4.5 Tear strength. Three specimens prepared as specified in 4.4.4.1 shall be tested in accordance with ASTM D624, using die C.

4.4.4,6 Brittleness. Five specimens prepared as specified in 4.4.4.1 shall be tested at -10° + 2°F in an air heat transfer medium in accordance with ASTM D2137.

4.4.4.7 Volatile loss. Three specimens prepared as specified in 4.4.4.1 shall be tested in accordance with method A of ASTM D1203 except that the char-coal shall be new and unused Fisher Scientific Company Activated Coconut Shell Charcoal Catalog No. 5-685, 6 X 14 mesh, as manufactured by Barnebey-Cheney.

4,4.4,8 Viscosity. The edgeseal material shall be thoroughly mixed by stirring for at least 1 minute immediately prior to drawing the specimen and performing the viscosity test. The viscosity shall be determined using a model LVF Brookfield Viscosimeter assembled with a number 4 spindle at 30 revolutions per minute in accordance with the directions supplied by the manufacturer of

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the instrument. The edgeseal material shall be tested at a temperature of 77° + 2°F. Material shall be at least 7 days old as measured from date of manufacture. Viscosimeter shall employ the spindle guard during measurements. Readings shall be taken after 1.5 minutes ± 10 seconds.

4.4.5 Acceptance/rejection criteria. If the edgeseal material fails to meet any of the tests specified in 4.4.3, the lot represented shall be rejected.

5* PACKAGING

5.1 Unit packing and packing, interplant shipment (see 6.4). The edgeseal material shall be unit packed and packed to prevent damage in shipment from the supply source to the first receiving activity. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation.

5.2 Marking. In conjunction with information provided in the contract or order, marking shall be in accordance with Fed. Std. No. 123 and shall include contract number, lot number and date produced.

6. NOTES

6.1 Intended use. Type I edgeseal material is intended for use as a sealing material in the manufacture of filter elements for M17/M17Al/M17A2 masks. Type II is for filter panels in the M2 infant protector.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Special marking if required (see 5.2).
(c) The type of edgeseal material required (see 1.2).
(d) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract,
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Batch. A batch is defined as the quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Interplant shipment. Packaging for interplant shipment is for supplies and materials that do not directly enter the military supply system* Typical interplant shipments are shipments from a vendor to a contractor or from a contractor to a military arsenal.

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6.5 Low migration plastisol.

(a) Providence D-3557 and D-3558 Low Migration Plastisol can be purchased from Providence Coatings and Chemicals Division, Whittaker Corporation, King Phillip Road, West Providence, RI 02914.

(b) Wooster A-466-2 and A-466-2A Low Migration Plastisols can be purcahsed from Whittaker Corporation, Wooster-Universal Division, P.O> Box 382) Wooster, OH 44691.

6.6 Filter elements. When specified in the contract or purchase order, the preproduction inspection of MIL-F-51425 may be used for acceptance of the compatibility preproduction provisions of this specification.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-51420A(EA)	EDGESEAL MATERIALS, CHEMICAL-BIOLOGICAL FILTER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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